

May 8

How to do calculations

Local coordinates $M^n \supset \mathcal{U} \xrightarrow{\text{open}} \mathbb{R}^n$ $\vec{x} = (x^1, \dots, x^n)$

$$\text{In } \mathcal{U}, \quad g_{ij} = g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right)$$

$$g \mapsto \nabla \mapsto Rm.$$

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = \sum_k \Gamma_{ij}^k \frac{\partial}{\partial x^k} \quad \text{Christoffel symbols}$$

(g^{ij}) inverse matrix of (g_{ij}) .

$$\text{Using } \left[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right] = 0.$$

Formula

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} \left(\frac{\partial}{\partial x^i} g_{jl} + \frac{\partial}{\partial x^j} g_{il} - \frac{\partial}{\partial x^l} g_{ij} \right).$$

Ricci flow:

$$\frac{\partial}{\partial t} g_{ij} = -2R_{ij}.$$

$g(t)$ solution to RF, $\nabla^{g(t)}$ L.C. conn of $g(t)$.

$\frac{\partial}{\partial t} \nabla^{g(t)}$ is a tensor $(1,2)$ -tensor
contravariant $\uparrow$ $\nwarrow$ covariant.

In local coords:

Lemma

$$\frac{\partial}{\partial t} \Gamma_{ij}^k = \frac{1}{2} g^{kl} \left(\nabla_i \frac{\partial}{\partial t} g_{jl} + \nabla_j \frac{\partial}{\partial t} g_{il} - \nabla_l \frac{\partial}{\partial t} g_{ij} \right)$$

where $\nabla_i = \nabla_{\partial/\partial x^i}$.

Let $\frac{\partial}{\partial t} g_{jl} = h_{jl}$.

$$\nabla_i h_{jl} = \nabla h \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^l} \right) = \left(\nabla_{\frac{\partial}{\partial x^i}} h \right) \left(\frac{\partial}{\partial x^j}, \frac{\partial}{\partial x^l} \right)$$

are the components of a $(0,3)$ -tensor.

Proof. Both sides represent tensors (of type $(1,2)$).
 $\exists$

So if we can verify this equation in some coordinate system, then it's true (in all coordinate systems).
 $\forall$

Let $x \in M^n$ be any point.

Let $\{x^i\}$ be geodesic normal coordinates centered at x .

Hence $\partial_i g_{jk}(x) = 0$, & in particular $\Gamma_{ij}^k(x) = 0$.

Now

$$\frac{\partial}{\partial t} \Gamma_{ij}^k(x) = \frac{1}{2} \left(\frac{\partial}{\partial t} g^{kl} \right) \left(\frac{\partial}{\partial x^i} g_{jl} + \frac{\partial}{\partial x^j} g_{il} - \frac{\partial}{\partial x^l} g_{ij} \right)(x)$$

$$+ \frac{1}{2} g^{kl} \left(\frac{\partial}{\partial x^i} \frac{\partial}{\partial t} g_{jl} + \frac{\partial}{\partial x^j} \frac{\partial}{\partial t} g_{il} - \frac{\partial}{\partial x^l} \frac{\partial}{\partial t} g_{ij} \right)(x)$$

$$= \frac{1}{2} g^{kl} \left(\nabla_i \frac{\partial}{\partial t} g_{jl} + \nabla_j \frac{\partial}{\partial t} g_{il} - \nabla_l \frac{\partial}{\partial t} g_{ij} \right)(x)$$

LHS & RHS are equal & components of tensors.

So this represents a tensor equality. $\square$

Corollary Under RF $\frac{\partial}{\partial t} g_{ij} = -2R_{ij}$.

$$\frac{\partial}{\partial t} \Gamma_{ij}^k = -g^{kl} (\nabla_i R_{jl} + \nabla_j R_{il} - \nabla_l R_{ij})$$

↑
one derivative
of g .

→ 1 time deriv.

curvature = 2 deriv. of g .

3 deriv. of g .

||

1 + 2.

↑ space derivative

Lemma

$$\frac{\partial}{\partial t} g^{ij} = -g^{ik} g^{jl} \frac{\partial}{\partial t} g_{kl}$$

Since $\frac{\partial}{\partial t} A^{-1} = -A^{-1} \cdot \frac{\partial}{\partial t} A \cdot A^{-1}$, because $\frac{\partial}{\partial t} (A \cdot A^{-1}) = \frac{\partial}{\partial t} I = 0$.

M^n orientable

volume form. $dV = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n$.

$$\frac{\partial}{\partial t} \det A = \det A \cdot \text{trace} (A^{-1} \frac{\partial}{\partial t} A).$$

Lemma.

$$\frac{\partial}{\partial t} dV = \frac{1}{2} g^{ij} \frac{\partial}{\partial t} g_{ij} dV.$$

So, for RF,

$$\frac{\partial}{\partial t} dV = -R dV$$

So for g, g^{-1}, dV, ∇

Riemann curvature tensor:

$$Rm(U, V)W = \nabla_U (\nabla_V W) - \nabla_V (\nabla_U W) - \nabla_{[U, V]} W.$$

is tensorial.

R_{ijk}^l defined by

$$Rm\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right) \frac{\partial}{\partial x^k} = \sum_{l=1}^n R_{ijk}^l \frac{\partial}{\partial x^l}.$$

$(1, 3)$ -tensor.

$\frac{\partial}{\partial t} Rm$ in local coordinates.

Lemma $\left[\frac{\partial}{\partial t} R_{ijk}^l = \nabla_i \left(\frac{\partial}{\partial t} \Gamma_{jk}^l \right) - \nabla_j \left(\frac{\partial}{\partial t} \Gamma_{ik}^l \right) \right]$

$$R_{jik}^l = -R_{ijk}^l$$

↑
we have a formula for this.

$\frac{\partial}{\partial t} Rm$
1 time deriv of 2 deriv of g .
(4)

$\nabla \cdot (3 \text{ deriv } g)$.
(4)

Proof. Let $x \in M^n$ be any point.

Let $\{x^i\}$ be geodesic normal coordinates centered at x^i .

Recall:

$$R_{ijk}^l = \underbrace{\partial_i \Gamma_{jk}^l} - \underbrace{\partial_j \Gamma_{ik}^l} + \underbrace{\Gamma_{ip}^l \Gamma_{jk}^p} - \underbrace{\Gamma_{jp}^l \Gamma_{ik}^p}.$$

$$\frac{\partial}{\partial t} (\Gamma * \Gamma) = \frac{\partial}{\partial t} \Gamma * \Gamma + \Gamma * \frac{\partial}{\partial t} \Gamma.$$

↑ being vague.

at center of geod. norm. coord.

Thus $\frac{\partial}{\partial t} R_{ijk}^l(x) = \partial_i \left(\frac{\partial}{\partial t} \Gamma_{jk}^l \right) - \partial_j \left(\frac{\partial}{\partial t} \Gamma_{ik}^l \right).$

$$= \nabla_i \left(\frac{\partial}{\partial t} \Gamma_{jk}^l \right) - \nabla_j \left(\frac{\partial}{\partial t} \Gamma_{ik}^l \right). \quad \square$$

Under RF, $\frac{\partial}{\partial t} \Gamma_{jk}^l = -g^{lm} (\nabla_j R_{km} + \nabla_k R_{jm} - \nabla_m R_{jk}).$

This is a bit complicated.

For the most part R_m is not that important in RF.

Simplification ∴ take trace.

1st trace. set $i=l$ & sum.

$$R_{jk} = R_{ijk}^i$$

Compute. $\frac{\partial}{\partial t} R_{jk} = \frac{\partial}{\partial t} R_{ijk}^i$

$$= \nabla_i \left(\frac{\partial}{\partial t} \Gamma_{jk}^i \right) - \nabla_j \left(\frac{\partial}{\partial t} \Gamma_{ik}^i \right).$$

where $\frac{\partial}{\partial t} \Gamma_{jk}^i = -g^{il} (\nabla_j R_{kl} + \nabla_k R_{jl} - \nabla_l R_{jk}).$

& $\frac{\partial}{\partial t} \Gamma_{ik}^i = -g^{il} (\nabla_i R_{kl} + \nabla_k R_{il} - \nabla_l R_{ik}).$

$$\boxed{\frac{\partial}{\partial t} R_{jk} = -g^{il} (\nabla_i \nabla_j R_{kl} + \nabla_i \nabla_k R_{jl} - \nabla_i \nabla_l R_{jk} - \nabla_j \nabla_k R_{il})}$$

not looking like a heat eqn. But we can make it that way.

However, let's simplify again ∴ trace.

Scalar curvature.

$$R = g^{jk} R_{jk}.$$

Now. Lemma. Under RF,

$$\boxed{\frac{\partial}{\partial t} R = \Delta R + 2|Ric|^2}$$

Proof. $\frac{\partial}{\partial t} R = \frac{\partial}{\partial t} (g^{jk} R_{jk})$

$$= \frac{\partial}{\partial t} g^{jk} \cdot R_{jk} + g^{jk} \frac{\partial}{\partial t} R_{jk}$$

$$= -g^{jl} g^{km} \underbrace{\frac{\partial}{\partial t} g_{lm}}^{-2R_{lm}} \cdot R_{jk}$$

$$+ g^{jk} \left(-g^{il} \left(\underbrace{\nabla_i \nabla_j R_{kl} + \nabla_i \nabla_k R_{jl} - \nabla_i \nabla_l R_{jk} - \nabla_j \nabla_k R_{il}}_{- \Delta R} \right) \right)$$

$$= 2|Ric|^2 - 2 \underbrace{\text{div}(\text{div}(Ric))}_{\frac{1}{2} \nabla R} + 2 \Delta R$$

$$= \Delta R + 2|Ric|^2 \quad \text{contracted 2nd Bianchi.}$$

Post mortem: $-\frac{1}{2} \Delta R$

$$+ g^{jk} \left(-g^{il} \left(\nabla_i \nabla_j R_{kl} + \nabla_i \nabla_k R_{jl} - \nabla_i \nabla_l R_{jk} - \nabla_j \nabla_k R_{il} \right) \right)$$

$$- g^{il} \nabla_i (g^{jk} \nabla_j R_{kl})$$

$$(\text{div}(Ric))_l$$

$$= \frac{1}{2} \nabla_l R$$

each.

$$+ g^{il} \nabla_i \nabla_l (g^{jk} R_{jk})$$

$$\underbrace{\Delta R}$$

$$-\frac{1}{2} g^{\mu\nu} \nabla_\mu \nabla_\nu R = -\frac{1}{2} \Delta R.$$